AgriGuardian AI: An SOA-Based Agentic Framework for Carbon-Aware Autonomous Irrigation in Paddy Cultivation

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***Abstract*—Paddy cultivation requires substantial water resources, often leading to over-irrigation, excessive pump energy usage, and extended inundation periods that significantly heighten methane emissions. Traditional irrigation relies on static schedules or isolated single-variable thresholds, failing to simultaneously balance soil moisture, power consumption, standing water duration, and upcoming rainfall. To address these multi-dimensional operational and environmental constraints, this paper introduces AgriGuardian AI, a Service-Oriented Architecture (SOA)-based agentic framework designed for carbon-aware autonomous irrigation under Alternate Wetting and Drying (AWD) practices. The framework aggregates real-time field data—including soil moisture, pump power usage, inundation time, and rainfall predictions—to compute a novel Carbon-Aware Irrigation Score (CAIS). An autonomous decision agent utilizes this score to dynamically execute or recommend three distinct operational actions: Irrigate Now, Delay Irrigation, or Skip Irrigation. Implemented through modular SOA microservices, AgriGuardian AI seamlessly links IoT-based field monitoring with high-level optimization logic. This adaptive framework eliminates non-productive water and energy consumption while actively preventing environmental conditions that facilitate methane generation in rice paddies.**

***Index Terms*—Carbon-Aware Irrigation, Agentic AI, Service-Oriented Architecture, Paddy Cultivation, Alternate Wetting and Drying, Smart Agriculture**

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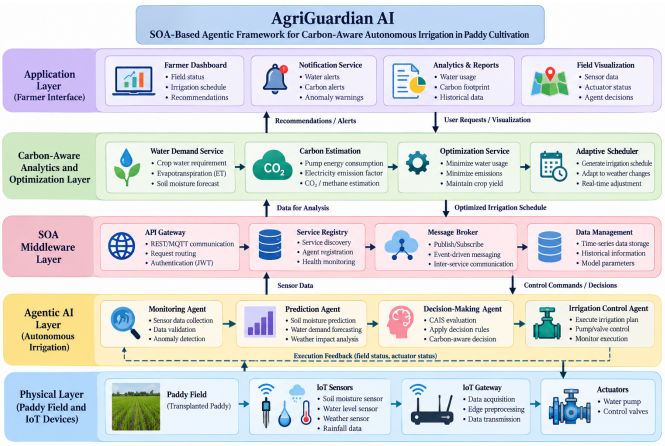


Fig. 1. Overall architecture of the proposed AgriGuardian AI framework for carbon-aware autonomous irrigation in paddy cultivation.

I. Introduction

Rice is a critical staple crop globally, but conventional paddy cultivation is exceptionally resource-intensive, requiring continuous field flooding to maintain optimal growth conditions. Standard manual operations and fixed irrigation schedules often lead to severe operational inefficiencies, including severe over-irrigation, inflated pump electricity consumption, and prolonged standing water conditions. Beyond water and energy

waste, continuous soil saturation creates anaerobic conditions in paddy fields, serving as a major contributor to agricultural greenhouse gas emissions, specifically methane (CH_4). Modern sustainable agricultural practices promote Alternate Wetting and Drying (AWD) as an effective management system to periodically cycle paddy fields between saturated and dry conditions. However, executing precise AWD in the field requires strictly maintaining optimum water depths (typically 20–40 mm) while continuously accounting for complex, interrelated environmental variables such as soil moisture, ambient temperature, crop growth phase, electricity availability, and local rainfall events.

- **Carbon-Aware Agentic Framework:** We propose an agentic system tailored for paddy AWD management that evaluates real-time field conditions, energy metrics, and rainfall forecasts to automate irrigation control.
- **Carbon-Aware Irrigation Score (CAIS):** We introduce a unified quantitative decision metric (CAIS) that jointly synthesizes soil moisture levels, pump power consumption, methane emission risk conditions, and weather forecasts to guide automated decision-making.
- **Dynamic Decision Logic:**The autonomous agent renders one of three precise operational actions—"Irrigate Now", "Delay Irrigation", or "Skip Irrigation"—optimizing water consumption and avoiding redundant energy usage when rainfall is anticipated.
- **Modular SOA Architecture:** We implement a scalable, service-oriented middleware structure that allows seamless integration of heterogeneous IoT sensors, external prediction APIs, and actuator control interfaces without disrupting core system functions.

  To improve the efficiency of irrigation, advances in Artificial Intelligence (AI) and autonomous decision-making allow intelligent control mechanisms to make decisions based on a number of different environmental and operational factors simultaneously. Modern conventional irrigation systems tend to offer only ON/OFF control of pump, yet AI-driven irrigation systems can take into account the soil/crop conditions weather water availability, energy consumption and environment when planning their irrigation action. Multi-factor decision-making like this is mostly desirable in carbon-aware agricultural systems as both energy required for pumping, and environment of aggressive flooding can be taken into consideration. To meet these requirements, this paper introduces an open service oriented architecture (SOA) based context-aware autonomous agentic system for: (1) carbon-conscious autonomous irrigation control in transplanted paddy cultivation, as shown in figure 2. The system offers aggregation of several supporting services to gather and process various sources of application-dependent irrigation information and make it available to an autonomous decision maker agent. These information includes soil moisture or water level, pump energy consumption, field temperature, daily high/low temperature, water inundating time and local weather forecast to set a reasonable irrigation time based on existing situation of paddy field.

  The primary achievement of this paper is the: a numeric value that informs whether control of irrigation is warranted. It takes into account necessary irrigation, soil, and water conditions, pump energy requirements, methane-emission risk, and other environmental issues. The autonomous decision agent then makes one of three decisions"Irrigate Now", "Postpone Irrigation", or "Skip Irrigation". If there is a prediction of rainfall, irrigation time can be postponed or avoided, saving water and energy.

  The individual modules of system are based on approach of Service-Oriented Architecture (SOA) where functions like sensing, data collection, data processing irrigation decision-making and interaction with users are organized as loosely coupled services, which communicate based on interfaces built among themselves. The independent module so can be managed independently and can extend the functionality. The autonomous agent combines data information from individual modules and thereby can analyze the conditions of the farm fields and environment and finally generate the irrigation advice. In short, AgriGuardian AI merges together the components of sensor-based sensing, autonomous decision-making by AI, CAIS-based carbon bookkeeping and SOA-based service integration in a single system.

TABLE I
COMPARISON OF EXISTING APPROACHES AND AGRIGUARDIAN AI

Aspect	AWD	Smart	Proposed
Paddy cultivation	Yes	Yes	Yes
AWD	Yes	Some	Yes
Soil/water monitoring	Yes	Yes	Yes
Rainfall forecast	Limited	Yes	Yes
Pump energy	Limited	Some	Yes
Flooding duration	Yes	Limited	Yes
Carbon-aware decision	Limited	Limited	Yes
Autonomous agent	Limited	Some	Yes
SOA architecture	Limited	Limited	Yes
CAIS metric	No	No	Yes
Irrigation actions	Threshold	Automated	3 actions

II. RELATED WORK

Awidely-focused area of research into sustainable rice irrigation is that concerned with saving water without causing any adverse impact on production, as well as reducing the impact of rice irrigation on the environment, mostly for green-houses gas emissions. Most of the research carried out providing insight suitable for water-control concepts like the proposedAgriGuardianAIcan generally be classified under these categories; Alternating Wetting and Drying(AWD), reducing greenhouse gases by water irrigationandintelligent irrigational decision-making. AWD is a well-documented water-application management technique designed to save water by minimizing

the extension of waterlogged conditions between phases of water-supply, like flooding during rice-growth. In principle, AWD encompasses three stages; thewetting-upstage in which water is added to the field, thedrying-downstage in which water application is eased off or stoppedand the rewetting-upstage in which further irrigation is introduced when necessary. These phases prevent considerable flooding, offering a potential reduction in water usage and mitigation of methane emissions. Because of this, current research is moving toward adopting water-saving procedures like AWD with intelligent decision making approaches in rice-irrigation.

A. Water Management and Greenhouse Gas Emissions

Water management is related with greenhouse gases emission from flooded fields. Continuous flooding can bring aerobic levels and methane production. With other way, AWD is marked by reduction of soil saturation level and, that means, it can increase emission of GHG. But, seems like the methane emission will be reduced; even nitrous oxide emission can be increased, based on soil, climate, and management conditions. Seems like a reduction of water use and GHG emission can be possibly achieved with AWD. Further research is necessary, but the decision of irrigation should not be based only on the water requirement of the crop, considering factors like moisture, duration of flooding rainfall temperature, energy consumption and GHG emissions together. Through a carbon-aware irrigation an economic model can be built, helping water management in paddy fields, considering water consumption and GHG emission simultaneously.

B. Intelligent Irrigation and Decision-Making

Actually, the recent trend in smart agriculture is to develop sensor-based automated and data-driven irrigation management system capable of accessing various indicators such as water level, water, and soil moisture, weather conditions rainfall flooding duration and energy consumption to determine the water requirement of crops and to schedule irrigations because of this. With the help of machine learning and predictive approaches, better irrigation scheduling can be achieved because of understanding the relationship between weather condition and crop water requirement. Though, most of the traditional automatic irrigation systems works on a single parameter or threshold. For instance, turn on irrigation when the soil moisture below a threshold level while disregarding the other important factors. Despite Really such systems are automated, they might miss other logical factors such as weather forecast and past conditions. But, advanced systems can take into account variables like rainfall in the previous 6, 12 and 24 hours, past flooding level and duration, electricity consumption, weather forecast and etc. Using them together with automatic control, more intelligent irrigation management can be achieved.

C. Research Gap

The progress of **water-saving irrigation management for rice production and greenhouse gases reduction by AWD' is remarkable in the literature. Under the existing research work of water savings and soil environmental effects in AWD, studies on correlation of threshold water depths with water saving, rice yield, methane, and other environmental effects have been carried out (Carrijo et al (Carrijo et al., 2017; Cheng et al., 2022; Gao et al., 2024). 2017; Cheng et al. 2022; Gao et al. 2024). Meanwhile, the intelligent irrigation system based on environmental information acquisition and automatic control has high potential to use the real-time environmental information and automated decision in water saving irrigation, but such work is rare in the literatures: researches on AWD has investigated only on water use, crop production and environmental effects, while intelligent irrigation systems were on automatic control with one parameter like soil moisture. Crucial parameters like **pump electricity consumption rain sustained flooding times, runoff, and total water consumption** are often ignored simultaneously. One contribution from AgriGuardian AI is to put these factors together in an **SOA-based agentic solution**. The proposed system quantifies these factors with the **Carbon-Aware Irrigation Score (CAIS)** being used as the decision criterion, jointly optimising irrigation needs, electricity consumption, and flooded scenario understanding. From the information it has, the agentic decision-maker should suggest the match of options of either "Irrigate Now, " "Delay Irrigation, " or "Skip Irrigation, " in particular when rain will occur.

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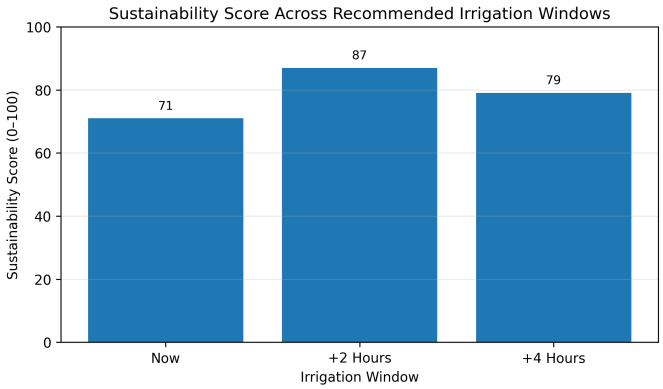


Fig. 2. Sustainability score across different irrigation windows.

III. Proposed System

AgriGuardian AI is a carbon-aware autonomous irrigation scheduling system for transplanted paddy using the AWD (Alternate Wetting and Drying) approach. It amalgamates several critical parameters of the field condition (i.e. soil moisture/water level, power consumption of the water pump, consecutive days of flooded field, and rainfall data) and gives to an autonomous decision agent, which observes the current field conditions corresponding parameters using the CAIS, and gives the final irrigation decision of "Irrigate Now", "Delay Irrigation", or "Skip irrigation" when rain is anticipated. Though the system checks multiple parameters of soil moisture and other environmental conditions, such multi-parameter based approach of soil moisture threshold based approach is more flexible and resource friendly for irrigation scheduling. The method of the proposed system employs a Service-Oriented architecture (SOA) in which collection and analysis of the data, and irrigation management are implemented as separate modular services, which are used by an autonomous decision agent to generate the final field condition.

The main components of the proposed system are:

- **Field Monitoring:** Collects soil moisture or water-level information from the paddy field.
- **Energy Monitoring:** Records the electricity consumption associated with irrigation pump operation.
- **Flooding Monitoring:** Tracks the number of continuous days for which the field remains flooded.
- **Weather Monitoring:** Provides rainfall forecast information for upcoming irrigation decisions.
- **Decision Agent:** Processes the available inputs and evaluates the irrigation condition using CAIS.
- **Irrigation Management:** Executes or recommends the appropriate irrigation action based on the agent's decision.

IV. SOA-Based Implementation

The AgriGuardian AI system is designed based on Service-Oriented Architecture (SOA) architecture in which the major functions are designed as various modular and independent services with each other through the well-designed interface. The infrastructure of this architecture contains various discrete services like **collecting field data, irrigationspecific information processing, field condition analyzing and irrigations recommendation generating service, etc. The independent decision agent consumes the information from those services and then makes the decision data shows in addition the Carbon-Aware Irrigation Score (CAIS) value.

The main workflow of the implementation is:

1) Field and environmental data are collected.
2) The collected data are processed by the corresponding services.
3) The processed information is provided to the decision agent.
4) The decision agent evaluates the Carbon-Aware Irrigation Score (CAIS).
5) An irrigation action is generated based on the evaluated conditions.

The resulting decision is one of three possible actions: *Irrigate Now*, *Delay Irrigation*, or *Skip Irrigation*. This service-based implementation allows the irrigation decision process to be extended with additional sensing, prediction, or decision services without changing the complete system architecture.

TABLE II
Inputs and Outputs of AgriGuardian AI

Category	Parameter
Field condition	Soil moisture / water level
Energy	Pump electricity consumption
Flooding	Continuous flooded days
Weather	Rainfall forecast
Decision	Irrigate Now
Decision	Delay Irrigation
Decision	Skip Irrigation

V. Results and Discussion

The proposed AgriGuardian AI system autonomously irrigates in response to the field water status, pump electricity consumption, soil information, current weather, and weather forecast. It is capable of using past basin information and current rainfall data to determine the importance of the forecasted rainfall, providing this information via services based on SOA. The decision agent is then given the set of information and generates one of three recommendations the "Irrigate Now, " "Delay Irrigation, " or "Skip Irrigation." The system adapts to current field and weather conditions via the use of set of services instead of an irrigation schedule. This system is first examined at the level of the service outputs, then through the full irrigation decision making flow, and finally how the CAIS combines the different set of inputs parameters. The SOA modular design allows each individual service to operate independently, while the decision agent processes that information to generate

irrigate recommendations. Quantitative testing under real field and sensor data needs to be done before conclusive results about water savings, energy savings, irrigation system efficiency, and environmental impact can be determined.

VI. Conclusion and Future Work

This paper proposes AgriGuardian AI, an SOA-based agentic system for carbon-aware autonomous irrigation management for transplanted paddy cultivation. Management solutions will optimize multiple considerations including soil moisture or water level, pump electricity cost, flooding duration, and rainfall prediction to enable flexible irrigation control. The paper proposed the Carbon-Aware Irrigation Score (CAIS)- a unified parameters by considering irrigated condition, energy consumption, and CO2 emission indicators as decision maker inputs. Based on the irrigated condition, the autonomous decision agent will suggest one action out of ”Irrigate Now, ” ”Delay Irrigation, ” and ”Skip Irrigation.” Modular SOA-based system offers a structure including data collection, processing, and decision-making services. Future work will include field tests through sensors data under relevant field cropping conditions and larger paddy-farming case datasets. Water usage, pump power consumption, CO2 emission performance, and effects of different irrigation conditions and equipment are to be quantitatively evaluated to provide practical recommendations. Field experiments with practical irrigation equipment can further test and validate the proposed AgriGuardian AI system.

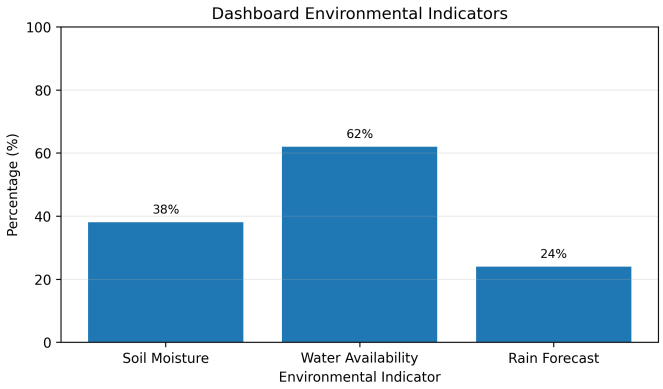


Fig. 3. Environmental indicators monitored by the AgriGuardian AI prototype.

VII. Implementation

The AgriGuardian AI dashboard provides a unified interface for monitoring paddy-field conditions and viewing autonomous irrigation decisions. It displays key parameters such as soil moisture, temperature, water availability, and carbon impact. The dashboard also provides an environmental-condition view, current irrigation decision, sustainability score, and AI agent status. These components allow the user to observe field conditions and the recommendations generated by the agentic. irrigation system.

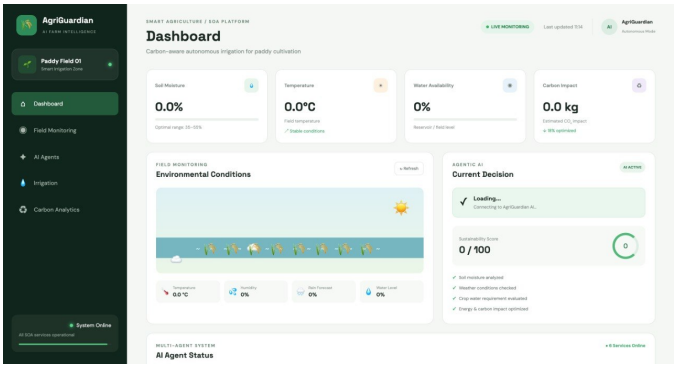


Fig. 4. AgriGuardian AI dashboard for field monitoring, autonomous irrigation decision-making, and carbon-aware analytics.

The irrigation control module provides an interface for monitoring and controlling the operation of the water pump. When irrigation is active, the interface displays the current pump status and indicates that water is being delivered to the paddy field. The module also provides a control option to stop irrigation when the required field condition has been achieved or when the decision agent determines that irrigation should be terminated.

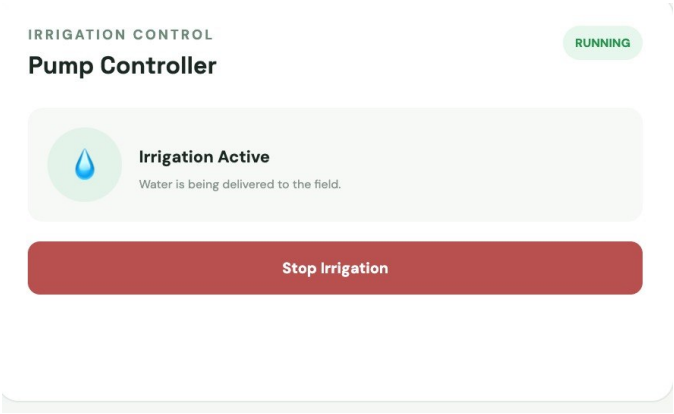


Fig. 5. Pump controller interface for irrigation control in the AgriGuardian AI system.

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